

## SAÉ2.04-3: Création d'une infrastructure AD avec Samba4

### Question 1 : Création du réseau Docker

Avec cette commande :

**docker network create --driver=bridge --subnet=172.20.0.0/16 --gateway=172.20.0.1 SAE204**

```
C:\Users\userlocal>docker network create --driver=bridge --subnet=172.20.0.0/16 --gateway=172.20.0.1 SAE204
79bfe485e280947e6effed86489ce38d0cd0623bfe844400a376c8312a77a50d
```

Vérification avec la commande :

docker network inspect SAE204

```
C:\Users\userlocal>docker network inspect SAE204
[
  {
    "Name": "SAE204",
    "Id": "79bfe485e280947e6effed86489ce38d0cd0623bfe844400a376c8312a77a50d",
    "Created": "2025-06-19T14:39:20.421354655Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.20.0.0/16",
          "Gateway": "172.20.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": {},
    "Options": {},
    "Labels": {}
  }
]
```

Question 2 Démarrez un conteneur sur ce réseau

`docker run -it --privileged --hostname samba-dc --add-host samba-dc:172.20.0.10 --network SAE204 -ip 172.20.0.10 ubuntu:latest bash`

```
C:\Users\userlocal\Documents\SAE204-3>docker run -it --privileged --hostname samba-dc --add-host samba-dc:172.20.0.10 --network SAE204 --ip 172.20.0.10 ubuntu:latest bash
```

Vérification

```
root@samba-dc:/# hostname
samba-dc
root@samba-dc:/# cat /etc/hosts
127.0.0.1        localhost
::1             localhost ip6-localhost ip6-loopback
fe00::          ip6-localnet
ff00::          ip6-mcastprefix
ff02::1         ip6-allnodes
ff02::2         ip6-allrouters
172.20.0.10     samba-dc
172.20.0.10     samba-dc
root@samba-dc:/# |
```

Question 3 Installez les paquets samba, smbclient et cifs-utils à votre conteneur

**Mise à jour des paquets :**

`docker exec -it ubuntu-serveur apt update`

```
C:\Users\userlocal\Documents\SAE204-3>docker exec -it ubuntu-serveur apt update
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1106 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [1509 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [1454 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1408 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1107 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1428 kB]
Fetched 8391 kB in 7s (1128 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
15 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

**Installation des paquets :**

`docker exec -it ubuntu-serveur apt install -y samba smbclient cifs-utils nano net-tools iputils-ping`

```
C:\Users\userlocal\Documents\SAE204-3>docker exec -it ubuntu-serveur apt install -y samba smbclient cifs-utils nano net-tools iputils-ping
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
samba is already the newest version (2:4.19.5+dfsg-4ubuntu9).
smbclient is already the newest version (2:4.19.5+dfsg-4ubuntu9).
Suggested packages:
  winbind bash-completion hunspell
The following NEW packages will be installed:
  iputils-ping libcap2-bin libpam-cap nano net-tools
The following packages will be upgraded:
  cifs-utils
1 upgraded, 5 newly installed, 0 to remove and 14 not upgraded.
Need to get 673 kB of archives.
After this operation, 1989 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libcap2-bin amd64 1:2.66-5ubuntu2.2 [34.2 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 iputils-ping amd64 3:20240117-1build1 [44.3 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libpam-cap amd64 1:2.66-5ubuntu2.2 [12.5 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 nano amd64 7.2-2ubuntu0.1 [282 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 cifs-utils amd64 2:7.0-2ubuntu0.2 [96.6 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 net-tools amd64 2.10-0.1ubuntu4.4 [204 kB]
Fetched 673 kB in 0s (1889 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package libcap2-bin.
(Reading database ... 8832 files and directories currently installed.)
Preparing to unpack .../0-libcap2-bin_1%3a2.66-5ubuntu2.2_amd64.deb ...
Unpacking libcap2-bin (1:2.66-5ubuntu2.2) ...
Selecting previously unselected package iputils-ping.
```

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```
Selecting previously unselected package net-tools.
Preparing to unpack .../5-net-tools_2.10-0.1ubuntu4.4_amd64.deb ...
Unpacking net-tools (2.10-0.1ubuntu4.4) ...
Setting up net-tools (2.10-0.1ubuntu4.4) ...
Setting up cifs-utils (2:7.0-2ubuntu0.2) ...
Setting up libcap2-bin (1:2.66-5ubuntu2.2) ...
Setting up nano (7.2-2ubuntu0.1) ...
update-alternatives: using /bin/nano to provide /usr/bin/editor (editor) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/editor.1.gz because associated file /usr/share/man/man1/nano.1.gz (of link group editor) doesn't exist
update-alternatives: using /bin/nano to provide /usr/bin/pico (pico) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/pico.1.gz because associated file /usr/share/man/man1/nano.1.gz (of link group pico) doesn't exist
Setting up libpam-cap:amd64 (1:2.66-5ubuntu2.2) ...
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/share/perl5/Debconf/FrontEnd/Dialog.pm line 79.)
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: (Can't locate Term/ReadLine.pm in @INC (you may need to install the Term::ReadLine module) (@INC entries check d: /etc/perl /usr/local/lib/x86_64-linux-gnu/perl/5.38.2 /usr/local/share/perl/5.38.2 /usr/lib/x86_64-linux-gnu/perl5/5.38 /usr/share/perl5 /usr/lib/x86_64-linux-gnu/perl-base /usr/lib/x86_64-linux-gnu/perl/5.38 /usr/share/perl/5.38 /usr/local/lib/site_perl) at /usr/share/perl5/Debconf/FrontEnd/Readline.pm line 8.)
debconf: falling back to frontend: Teletype
Setting up iputils-ping (3:20240117-1build1) ...
```

## Vérification

docker exec -it ubuntu-serveur smbdc --version

```
C:\Users\userlocal\Documents\SAE204-3>docker exec -it ubuntu-serveur smbdc --version
Version 4.19.5-Ubuntu
```

Question 4 Utilisez les commandes vues dans le TP1 de la ressource R1.01c afin d'approvisionner le serveur Samba ADDC

```
rm /etc/samba/smb.conf && samba-tool domain provision --use-rfc2307 --realm=IUT.LOCAL --domain=IUT --adminpass=Pa$$w0rd --server-role=dc
```

```

root@samba-dc:/# rm /etc/samba/smb.conf
root@samba-dc:/# samba-tool domain provision --use-rfc2307 --realm=IUT.LOCAL --domain=IUT --adminpass=Pa$$w0rd --server-
role=dc
INFO 2025-06-19 18:34:43,589 pid:294 /usr/lib/python3/dist-packages/samba/provision/__init__.py #2128: Looking up IPv4 a
ddresses
INFO 2025-06-19 18:34:43,591 pid:294 /usr/lib/python3/dist-packages/samba/provision/__init__.py #2145: Looking up IPv6 a
ddresses
WARNING 2025-06-19 18:34:43,591 pid:294 /usr/lib/python3/dist-packages/samba/provision/__init__.py #2152: No IPv6 addres
s will be assigned
INFO 2025-06-19 18:34:43,974 pid:294 /usr/lib/python3/dist-packages/samba/provision/__init__.py #2318: Setting up share.
ldb
INFO 2025-06-19 18:34:44,039 pid:294 /usr/lib/python3/dist-packages/samba/provision/__init__.py #2322: Setting up secret
s.ldb
INFO 2025-06-19 18:34:44,082 pid:294 /usr/lib/python3/dist-packages/samba/provision/__init__.py #2327: Setting up the re
gistry
INFO 2025-06-19 18:34:44,252 pid:294 /usr/lib/python3/dist-packages/samba/provision/__init__.py #2330: Setting up the pr
ivileges database
INFO 2025-06-19 18:34:44,335 pid:294 /usr/lib/python3/dist-packages/samba/provision/__init__.py #2333: Setting up idmap
db
INFO 2025-06-19 18:34:44,392 pid:294 /usr/lib/python3/dist-packages/samba/provision/__init__.py #2340: Setting up SAM db
INFO 2025-06-19 18:34:44,410 pid:294 /usr/lib/python3/dist-packages/samba/provision/__init__.py #886: Setting up sam.ldb
partitions and settings
INFO 2025-06-19 18:34:44,411 pid:294 /usr/lib/python3/dist-packages/samba/provision/__init__.py #898: Setting up sam.ldb
rootDSE
INFO 2025-06-19 18:34:44,422 pid:294 /usr/lib/python3/dist-packages/samba/provision/__init__.py #1320: Pre-loading the S
amba 4 and AD schema
Unable to determine the DomainSID, can not enforce uniqueness constraint on local domainSIDs

```

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## Verification

```

S-1-5-21-2298943957-4070809117-894059284
root@samba-dc:/# smbdc --foreground --debug-stdout
smbd version 4.19.5-Ubuntu started.
Copyright Andrew Tridgell and the Samba Team 1992-2023
main: server role = 'active directory domain controller' not compatible with running smbd standalone.
You should start 'samba' instead, and it will control starting smbd if required
root@samba-dc:/# samba -i
samba version 4.19.5-Ubuntu started.
Copyright Andrew Tridgell and the Samba Team 1992-2023
Attempting to autogenerate TLS self-signed keys for https for hostname 'SAMBA-DC.iut.local'
/usr/sbin/winbindd: Failed to exec child - No such file or directory
winbindd daemon died with exit status 255
task_server_terminate: task_server_terminate: [winbindd child process exited]
samba_terminate: samba_terminate of samba 297: winbindd child process exited
root@samba-dc:/# TLS self-signed keys generated OK

root@samba-dc:/# smbstatus

Samba version 4.19.5-Ubuntu
PID Username Group Machine Protocol Version Encryption Signing
-----

```

| Service | pid | Machine | Connected at | Encryption | Signing |
|---------|-----|---------|--------------|------------|---------|
| -----   |     |         |              |            |         |

## Question 5 : Création des groupes

samba-tool group add direction

samba-tool group add administration

samba-tool group add production

```

This is normal if an SMB client has never connected to your server.
root@samba-dc:/# samba-tool group add direction && samba-tool group add administration && samba-tool group add productio
n
Added group direction
Added group administration
Added group production

```

## Vérification

samba-tool group list | grep -E  
"direction|administration|production"

```
root@samba-dc:/# samba-tool group list | grep -E "direction|administration|production"
administration
direction
production
```

## Question 6 : Création des comptes utilisateurs

```bash

samba-tool user create ceo --given-name=CEO --surname=Director --  
mail-address=ceo@domain.org --login-shell=/bin/bash

samba-tool user create cfo --given-name=CFO --surname=Financial --  
mail-address=cfo@domain.org --login-shell=/bin/bash

samba-tool user create cto --given-name=CTO --surname=Technical --  
mail-address=cto@domain.org --login-shell=/bin/bash

```
root@samba-dc:/# samba-tool user create ceo --given-name=CEO --surname=Director --mail-address=ceo@domain.org --login-shell=/bin/bash
New Password:
Retype Password:
User 'ceo' added successfully
root@samba-dc:/# samba-tool user create cfo --given-name=CFO --surname=Financial --mail-address=cfo@domain.org --login-shell=/bin/bash
New Password:
Retype Password:
User 'cfo' added successfully
root@samba-dc:/# samba-tool user create cto --given-name=CTO --surname=Technical --mail-address=cto@domain.org --login-shell=/bin/bash
New Password:
Retype Password:
User 'cto' added successfully
```

## Vérification

samba-tool user list | grep -E "ceo|cfo|cto"

```
root@samba-dc:/# samba-tool user list | grep -E "ceo|cfo|cto"
ceo
cfo
cto
```

Question 7 : Ajout aux groupe

samba-tool group addmembers direction ceo

samba-tool group addmembers administration cfo

samba-tool group addmembers production cto

```
root@samba-dc:/# samba-tool group addmembers direction ceo
Added members to group direction
root@samba-dc:/# samba-tool group addmembers administration cfo
Added members to group administration
root@samba-dc:/# samba-tool group addmembers production cto
Added members to group production
```

Vérification

samba-tool group listmembers direction # Doit afficher "ceo"

samba-tool group listmembers administration # Doit afficher "cfo"

samba-tool group listmembers production # Doit afficher "cto"

```
root@samba-dc:/# samba-tool group listmembers direction && \
samba-tool group listmembers administration && \
samba-tool group listmembers production
ceo
cfo
cto
root@samba-dc:/#
```

Question 8 : Configuration Samba

```
root@samba-dc:/# cat <<EOF > /etc/samba/smb.conf
[global]
    workgroup = IUT
    security = user
    username map = /etc/samba/smbusers

[dir]
    path = /partage/dir
    valid users = @direction
    read only = no
    write list = @direction

[adm]
    path = /partage/adm
    valid users = @administration
    read only = no
    write list = @administration

[pro]
    path = /partage/pro
    valid users = @production
    read only = no
    write list = @production

[public]
    path = /partage/public
    guest ok = yes
    read only = no
    force user = nobody

[dir-adm]
    path = /partage/dir-adm
    valid users = @direction,@administration
    read only = yes
EOF write list = ceodirection,@production
```

Vérifier :

testparm -s

```
root@samba-uc:/# testparm -s
Load smb config files from /etc/samba/smb.conf
Loaded services file OK.
Weak crypto is allowed by GnuTLS (e.g. NTLM as a compatibility fallback)

Server role: ROLE_STANDALONE

# Global parameters
[global]
    security = USER
    username map = /etc/samba/smbusers
    workgroup = IUT
    idmap config * : backend = tdb

[dir]
    path = /partage/dir
    read only = No
    valid users = @direction
    write list = @direction

[adm]
    path = /partage/adm
    read only = No
    valid users = @administration
    write list = @administration

[pro]
    path = /partage/pro
    read only = No
    valid users = @production
    write list = @production
```



[public]

force user = nobody  
guest ok = Yes  
path = /partage/public  
read only = No

[dir-adm]

path = /partage/dir-adm  
valid users = @direction @administration  
write list = @direction

[dir-pro]

path = /partage/dir-pro  
valid users = @direction @production  
write list = ceo